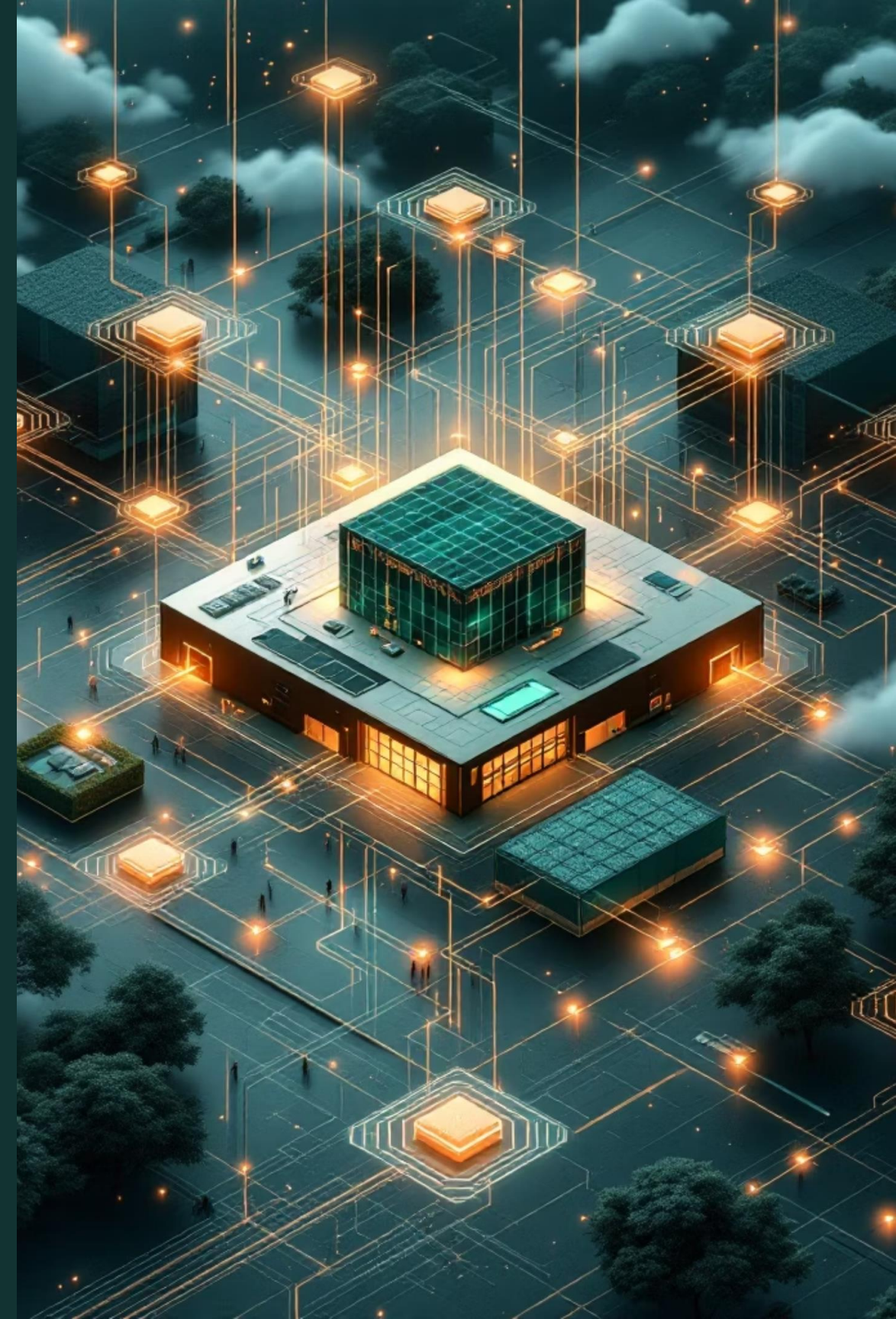


Integrated Customer Item Rating Warehouse

FudgeMart & FudgeFlix — IST 722 Group Project

IST 722

DATA WAREHOUSING



Project Overview

Two companies in the process of merging — each with its own customer feedback system — present a unique data integration challenge. Our project bridges that gap by building a unified analytical platform.

FudgeMart

A retail operation where customers purchase physical products and leave star ratings after their transactions.

FudgeFlix

A streaming service where subscribers rate titles after viewing — films, series, and other digital content — in a separate, siloed system.

-   Goal: Design and implement a unified data warehouse that consolidates customer rating data from both companies, enabling cross-company satisfaction analysis and strategic decision-making.



Why Integration Matters

When two companies merge, their data assets must merge too. Siloed feedback systems prevent leadership from seeing the full picture — and that blind spot has real business consequences.



Performance Clarity

Which offerings — products or titles — are truly top-rated across the combined portfolio?



Satisfaction Drivers

What factors most influence a customer's rating? Are patterns consistent across both business units?



Cross-Company Benchmarking

How do physical product ratings compare to streaming title ratings?
What can each business learn from the other?



Strategic Investment

Where should the newly merged organization direct resources to maximize customer satisfaction and retention?

Without integration: siloed data creates limited, fragmented decision-making. **With integration:** unified insights drive confident, evidence-based strategy.

Project Goals & Deliverables

Project Goals

- **Integrate** customer rating data from FudgeMart and FudgeFlix into a single analytical model
- **Standardize** feedback structures so that reviews and ratings share a common schema regardless of source
- **Enable** cross-company analytics – comparing satisfaction across retail products and streaming titles
- **Support** category-level analysis to identify high- and low-performing segments

Deliverables

✓ **Integrated dimensional model** – a cohesive design spanning both source systems

✓ **Star schema design** – optimized for BI tools and analytical queries

✓ **Integration documentation** – full write-up of design decisions and modeling rationale

BUSINESS PROCESS

The Shared Business Process: Customer Item Rating

The foundation of any dimensional model is a clearly defined business process. For this project, we identified **customer item rating** as the single process shared by both FudgeMart and FudgeFlix — the event where a customer evaluates something they consumed.

FudgeMart

A customer purchases a physical product → completes a review → submits a star rating. The rating is anchored to the product and the purchase transaction.

FudgeFlix

A subscriber watches a streaming title → engages with the content → submits a numeric or star-based rating. The rating is anchored to the title and the viewing session.

👉 This **shared behavioral pattern** — customer evaluates item after consumption — is the conceptual bridge that makes a unified warehouse model both logical and achievable.

Grain of the Data

One Row Per Rating Event

Each record in the fact table represents a single, atomic rating submitted by one customer for one item on one date.

What Each Record Captures

1

One Customer

The individual who submitted the rating, regardless of which source system they belong to

2

One Item

The specific product or streaming title being evaluated

3

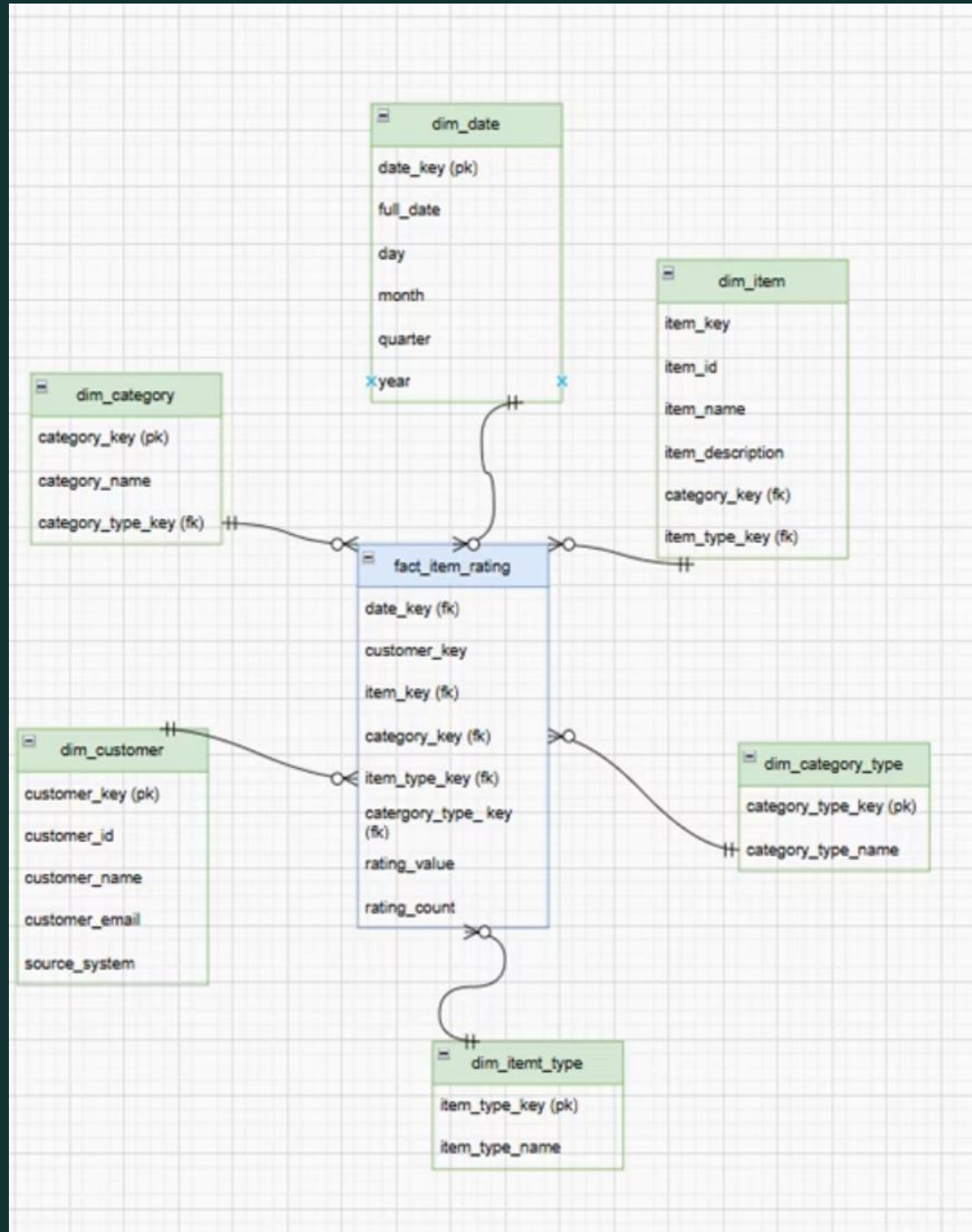
One Date

The calendar date on which the rating event occurred



Defining the grain at this level of atomicity ensures **consistent aggregation** across both businesses and prevents double-counting or ambiguity in analytical queries.

Star Schema Overview



Design Philosophy

We implemented a **pure star schema** — a single central fact table surrounded by denormalized dimension tables. This design minimizes query complexity and maximizes performance for BI reporting tools.

Central Fact Table

fact_item_rating — captures every rating event across both companies

Connected Dimensions

👤 dim_customer

📦 dim_item

🏠 dim_category

📌 dim_category_type

📅 dim_item_type

📅 dim_date

💡 This structure supports fast, flexible BI reporting with minimal joins — ideal for cross-company analysis.

Dimensions Deep Dive

Each dimension table plays a specific role in contextualizing rating events. Together, they provide a complete analytical lens across customers, items, categories, types, and time.



dim_customer

Represents every individual who submitted a rating. Includes customer ID, name, email, and – critically – a **source system attribute** identifying whether the customer originated from FudgeMart or FudgeFlix.



dim_item

A **unified item dimension** covering both physical retail products and streaming titles. Stores item name, description, and foreign keys to category and item type. Enables true cross-company item analysis.



dim_category & dim_category_type

FudgeMart uses **departments**; FudgeFlix uses **genres**. The category dimension stores the grouping, while category type identifies the classification system – preserving meaning while enabling unified queries.



dim_date & dim_item_type

dim_date provides full time context – calendar date, month, quarter, year, weekday – supporting trend and seasonal analysis. **dim_item_type** distinguishes physical products from streaming titles for cross-channel comparisons.

Integration Challenges & Modeling Solutions

Key Challenges Encountered

Physical vs. Digital Offerings

Retail products and streaming titles are fundamentally different entities requiring a flexible unified schema.

Conflicting Classification Systems

FudgeMart organizes by departments; FudgeFlix organizes by genres – incompatible hierarchies that needed reconciliation.

Separate Customer Databases

Each company maintained its own customer records with independent IDs and no common unique identifier.

Different Rating Methods

Review formats, scoring scales, and submission contexts varied between source systems.

Modeling Solutions Applied

01

Conformed Dimensions

Shared dimension tables (date, item type) that carry consistent meaning across both source systems

02

Surrogate Keys

System-generated integer keys replace source-system IDs, eliminating conflicts between FudgeMart and FudgeFlix records

03

Unified Item Structure

A single dim_item table accommodates both products and titles via the item_type foreign key

04

Category Type Distinction

Separating category and category_type preserves the semantic differences between departments and genres

05

Standardized Rating Measures

All ratings normalized to a consistent numeric scale stored in the fact table as rating_value and rating_count

Data Lineage & Implementation (dbt)

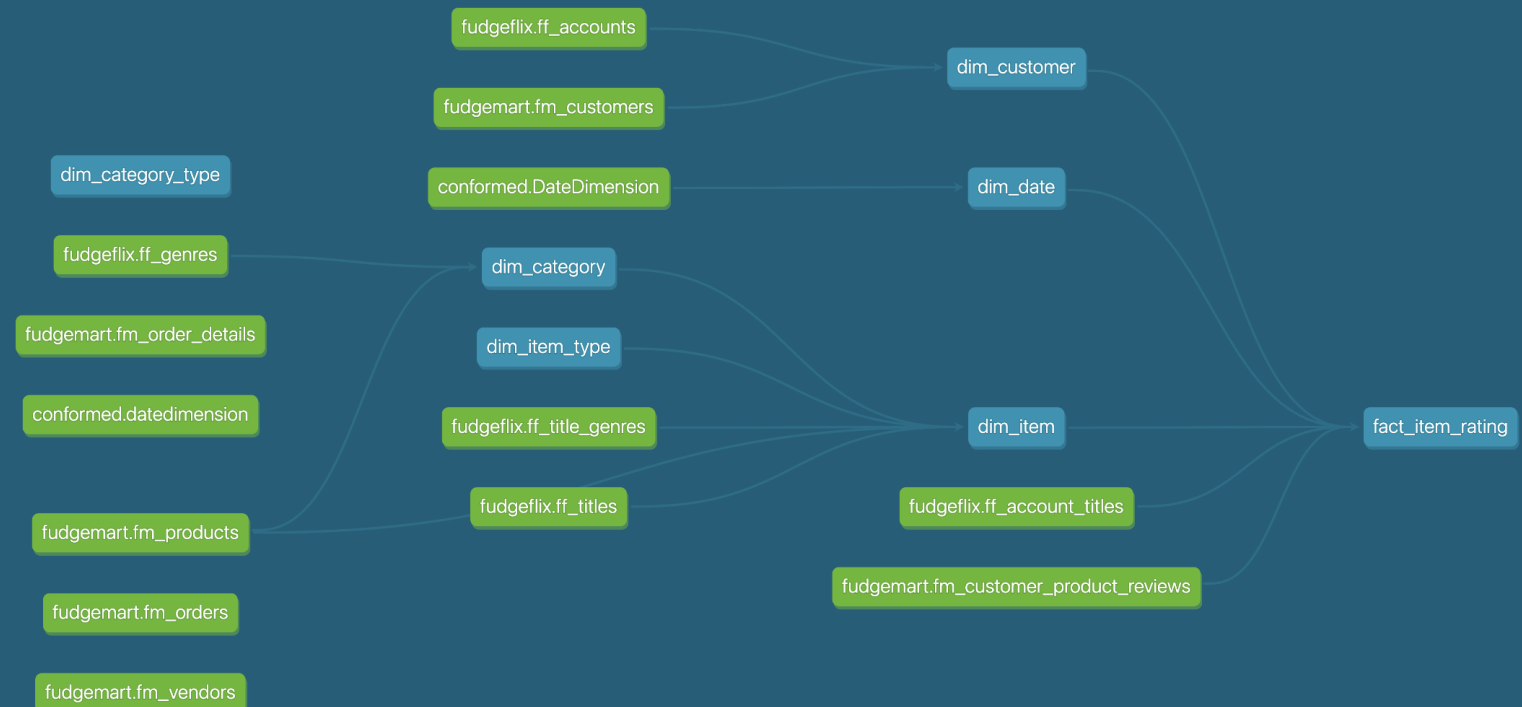
Our technical implementation utilized **dbt** to transform raw source data into the final dimensional model

Lineage Graph: The fact_item_rating table is downstream from unified dimensions, which themselves aggregate data from fudgemart and fudgeflix sources.

Surrogate Keys: To eliminate ID conflicts between the two To eliminate ID conflicts between the two companies, we used dbt_utils.generate_surrogate_key on a combination of source IDs and the source_system attribute.

Unified Transformations
SQL models utilize UNION ALL to combine records, such as merging physical products and streaming titles into a single dim_item structure.

Lineage Graph



```

1  {{ config(materialized='table') }}
2
3  WITH fudgemart_items AS (
4      SELECT
5          CAST(p.product_id AS VARCHAR) AS item_id,
6          p.product_name AS item_name,
7          p.product_description AS item_description,
8          'fudgemart' AS source_system,
9          c.category_key,
10         it.item_type_key,
11         ROW_NUMBER() OVER (PARTITION BY p.product_id ORDER BY p.product_id) AS rn
12     FROM {{ source('fudgemart', 'fm_products') }} p
13     LEFT JOIN {{ ref('dim_category') }} c
14         ON p.product_department = c.category_name
15     LEFT JOIN {{ ref('dim_item_type') }} it
16         ON 'product' = it.item_type
17 ),
18
19 fudgeflix_items AS (
20     SELECT
21         CAST(t.title_id AS VARCHAR) AS item_id,
22         t.title_name AS item_name,
23         t.title_synopsis AS item_description,
24         'fudgeflix' AS source_system,
25         c.category_key,
26         it.item_type_key,
27         ROW_NUMBER() OVER (PARTITION BY t.title_id ORDER BY tg.tg_genre_name) AS rn
28     FROM {{ source('fudgeflix', 'ff_titles') }} t
29     LEFT JOIN {{ source('fudgeflix', 'ff_title_genres') }} tg
30         ON t.title_id = tg.tg_title_id
31     LEFT JOIN {{ ref('dim_category') }} c
32         ON tg.tg_genre_name = c.category_name
33     LEFT JOIN {{ ref('dim_item_type') }} it
34         ON 'streaming_title' = it.item_type
35 )
36
37 SELECT
38     {{ dbt_utils.generate_surrogate_key(['item_id', 'source_system']) }} AS item_key,
39     item_id,
40     item_name,
41     item_description,
42     source_system,
43     category_key,
44     item_type_key
45 FROM fudgemart_items
46 WHERE rn = 1
47
48 UNION ALL
49
50 SELECT
51     {{ dbt_utils.generate_surrogate_key(['item_id', 'source_system']) }} AS item_key,
52     item_id,
53     item_name,
54     item_description,

```

ist722dbt > FinalProject > models >

dim_customer.sql

Save

```

1  {{ config(materialized='table') }}
2
3  WITH customers AS (
4
5      SELECT
6          customer_id AS customer_id,
7          customer_firstname AS first_name,
8          customer_lastname AS last_name,
9          customer_email AS customer_email,
10         NULL AS customer_segment,
11         'fudgemart' AS source_system
12     FROM {{ source('fudgemart', 'fm_customers') }}
13
14     UNION ALL
15
16     SELECT
17         account_id AS customer_id,
18         account_firstname AS first_name,
19         account_lastname AS last_name,
20         account_email AS customer_email,
21         NULL AS customer_segment,
22         'fudgeflix' AS source_system
23     FROM {{ source('fudgeflix', 'ff_accounts') }}
24
25 )
26
27 SELECT
28     {{ dbt_utils.generate_surrogate_key(['customer_id', 'source_system']) }} AS
29     *
30 FROM customers
31

```

</>

Commands

Code quality

Results

Compiled code

Lineage

2+dim_customer+2

Update Graph

SRC

fudgemart.fm_customers

SRC

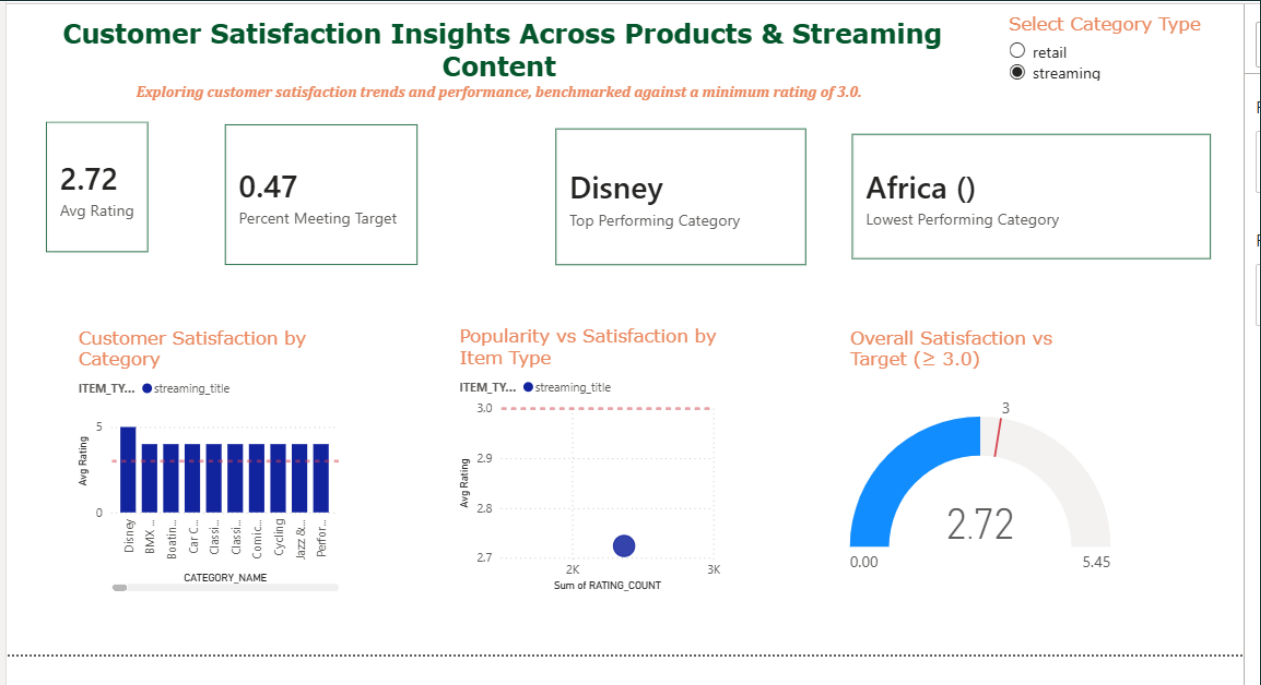
fudgeflix.ff_accounts

MDL

dim_customer

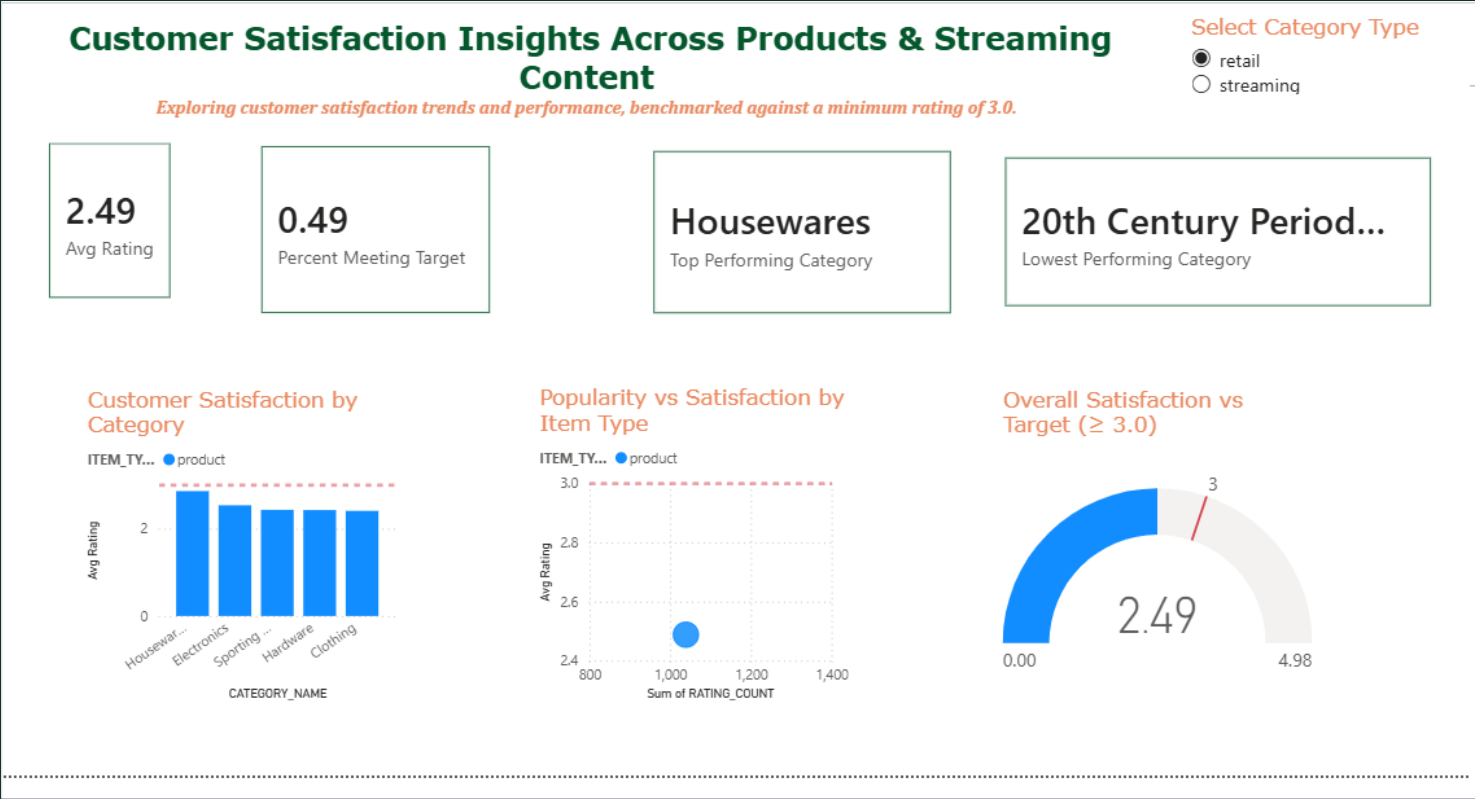
MDL

fact_item_rating



These interactive dashboards represent the final analytical layer of our integrated warehouse, powered by a **Snowflake** data backend and visualized through **BI reporting tools**.

By connecting our conformed dimensional models directly to this visualization layer, we provide leadership with a real-time, unified view of customer satisfaction across both the FudgeMart and FudgeFlix portfolios.



Performance Insights Across Retail and Streaming

Benchmark results

Streaming Content (FudgeFlix)

The streaming division maintains an average rating of **2.72**, with the "Disney" category leading performance and "Africa" trailing as the lowest-rated segment.

Retail Product (FudgeMart)

Physical products show a lower overall average rating of **2.49** "Housewares" is the top-performing category, whereas "20th Century Period..." is the lowest.

Gap to Quality Target (less than 3.0)

Both business units currently fall below the strategic satisfaction benchmark (3.0), with less than approximately 50% meeting the target rating.

Comparative Popularity vs Satisfaction

Integrated scatter plots reveal that streaming titles generally cluster at higher satisfaction levels compared to retail products, despite similar volume(rating count) distributions

Actionable Insight: By unifying these metrics, the organization can identify high performing segments (like Housewares and Disney) to replicate success factors in low performing areas.

Recommendations

To further enhance and improve customer experiences, we have collected some actionable recommendations that can be completed to improve the quality of the products and services.

1. Close the "Quality Gap" for Retail Products

With a current average rating of 2.49 and only 49% of products meeting the 3.0 target, the organization should launch a quality audit specifically for the "20th Century Period..." category. Identifying why this segment is underperforming relative to "Housewares" (the top retail performer) will help stabilize overall customer satisfaction.

2. Implement Cross-Platform Loyalty Incentives

Since streaming titles (2.72) outperform physical products (2.49), use the unified dim_customer data to identify high-value FudgeFlix subscribers and offer them targeted promotions for FudgeMart retail items. This leverages the higher streaming engagement to drive traffic toward the retail division.

4. Optimize Underperforming "Low-Volume" Segments

The "Africa" streaming category is currently the lowest-performing segment. Management should use the "Popularity vs. Satisfaction" scatter plot to determine if this is due to a lack of content variety or poor content quality, then either re-curate the category or reallocate that budget to higher-performing genres.

Team & Conclusion

Insights Enabled by the Warehouse

★ Top-rated items across both companies

📊 Cross-category satisfaction comparisons

📺 Product vs. streaming title performance

📈 Customer satisfaction trends over time

💡 Data-driven investment decisions

Team Contributions

Elizabeth

Business context, dimensional modeling, and requirements definition

Najah

Data pipeline design, dbt transformations, and Snowflake implementation

Sandra

Visualization, analytics layer, and final reporting

Conclusion

By centering our design on a shared business process — the moment a customer rates an item — we built a scalable, flexible analytical framework ready to serve the merged organization's strategic needs.

The warehouse transforms **siloes, incompatible feedback data** from two separate companies into a **unified source of actionable insight**. Leadership can now ask — and answer — questions that were simply impossible before integration.

📄 🧡 IST 722 — **Thank you!** We welcome your questions about our design decisions, integration approach, and implementation details.